EEE 143: Switching Theory and Digital Logic Design

2nd sem AY 2024-2025

Course Description: Fundamental concepts in Boolean Algebra for analyzing and designing logic circuits using Boolean expressions, logic gates, digital integrated circuit building blocks, and hardware description language.

Prerequisite: None

Course Credit: 3 units of lecture

Course Goals: To introduce techniques in analyzing digital logic circuits. To provide students with skills in designing combinational and sequential circuits using flip-flops, logic gates, simple digital building blocks, and hardware description language.

Mode of delivery: Mainly face-to-face lectures and problem-solving sessions.

Course outline:

Week	Topic	Hours
0 (Jan 22-24)	0.0 Class Policies 1.0 Introduction to Digital Systems	1.5
1 (Jan 29-31)	1.1 Number Systems 1.2 Integers and floating point numbers	1.5
2 (Feb 5-7)	2.0 Boolean Algebra 2.1 Set theory and Boolean logic 2.2 Logic gates 2.3 Combinational circuits, truth tables 2.4 Canonical forms	3
3 (Feb 12-14)	2.5 Boolean algebra 2.6 Minimization by Algebraic Manipulation	3
4 (Feb 19-21)	2.7 Karnaugh maps and logical adjacency 2.8 Incompletely specified functions	3
5 (Feb 26-28)	3.0 Combinational building blocks 3.1 Computer arithmetic 3.2 Half / Full Adder	3
6 (Mar 5-7)	3.3 Data selectors 3.4 Encoders and decoders	3
7 (Mar 12-14)	4.0 Combinational logic delay	1.5
Long Exam 1: March 15, 2025 (1-4 PM)		

8 (Mar 19-21)	5.0 Sequential circuits 5.1 Introduction to Sequential Circuits 5.2 Memory elements and excitation functions 5.3 State diagrams 5.4 Timing diagram of sequential circuits	3
9 (Mar 26-28)	5.5 Analysis of synchronous sequential circuits	3
10 (Apr 2-4)	5.6 State minimization	3
11 (Apr 9-11)	5.7 Synthesis of synchronous sequential circuits	1.5
Lenten Break		
12 (Apr 22-24)	5.8 Alternative synthesis techniques	3
13 (Apr 30-May 2)	6.0 Sequential circuit non-idealities 6.1 Flipflop non-idealities 6.2 Critical path delays	3
14 (May 7-9)	7.0 Special Topics 7.1 Algorithmic State Machines 7.2 Hardware description language	3
15 (May 14-16)	7.3 Towards processor design	3
Long Exam 2: May 24, 2025 (1-4 PM)		

References:

- Morris R. Mano, Charles R. Kime, Tom Martin. “Logic & Computer Design Fundamentals, 5th Edition (2016)”. Search for book through "e-Reserve Books" in UP Engineering Library Remote Access Portal (https://ezproxy.engglib.upd.edu.ph)
  https://upengglib.vstbridge.com/#/book-details/9781292096124
  Virtual e-reserve, can be borrowed for 14 days
- S. Brown and Z. Vranesic. “Fundamentals of Digital Logic with Verilog Design”, 3rd ed., Mc-Graw Hill, 2014.
- B. Lameres, “Introduction to Logic Circuits & Logic Design with Verilog”, Springer, 2019.
- M Mano and M. Ciletti, “Digital Design: With an Introduction to the Verilog HDL, VHDL, and SystemVerilog”, 6th Ed., 2018.
- D. Harris and S. Harris “Digital Design and Computer Architecture”, 2nd ed., Morgan Kaufmann, 2013.

Requirements and Grading:

There will be 1 online quiz or homework every week (and it will be due Friday of the following week). There will also be 2 face-to-face long exams, scheduled on Saturdays, 1-4PM. A student must have non-zero scores on all long exams. In case of an excused absence during an exam, a make-up exam will be given during finals week (schedule to be determined). The breakdown of your grade is shown below:

30% Weekly Quiz/Homework
60% Long exams
10% Class Participation

The following grading scale will be used:

92 – 100	**1.0**	72 – <76	**2.25**
88 – <92	**1.25**	68 – <72	**2.5**
84 – <88	**1.5**	64 – <68	**2.75**
80 – <84	**1.75**	60 – <64	**3.0**
76 – <80	**2.0**	<60	**5.0**

Lecture Instructors:

Isabel M. Austria
Email: isabel.austria@eee.upd.edu.ph

Adelson Chua
Email: adelson.chua@eee.upd.edu.ph

Lawrence Roman Quizon
Email: lawrence.quizon@eee.upd.edu.ph